

A Secure and Scalable DNA Data Storage Pipeline for Archival Big Data

Design, End-to-End Implementation, and Evaluation

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The Cold Data Dilemma & The Physical Storage Wall

Exponential Growth vs. Physical Scaling

- Over 80% of global Big Data represents **cold, archival records** requiring retention over decades.
- Conventional media face thermodynamic and physical barriers:
 - **HDD:** Superparamagnetic limit ($< 10 \text{ Tb/in}^2$); HAMR/BPM scaling delayed.
 - **NAND Flash:** Cell reliability degradation past 64 3D-stacked layers.
- High recurring operational expenditures: replacement every 5–7 years, ongoing cooling and power footprint.

DNA Data Storage (DDS)

- **Theoretical Density:** $\sim 455 \text{ EB/g}$ (2 bits/nucleotide).
- **Exceptional Longevity:** Half-life exceeding 500 years in silica.
- **Zero Idle Power:** Sustained preservation without power or maintenance.

Operational Bottleneck

Ultra-low write throughput ($\approx 400 \text{ B/s}$) and synthesis expense require high-efficiency encoding.

Problem Statement: The Security Blindspot in DNA Storage

State-of-the-Art Limitations

- State-of-the-art simulators (e.g., **DNACloud**) successfully address:
 - Base-3 ternary Huffman mapping.
 - Overlapping chunking for redundancy.
 - Biochemical parameter estimation (GC-content, melting temperature T_m).
- **Critical Limitation:** Data is treated as plaintext throughout the mapping process.

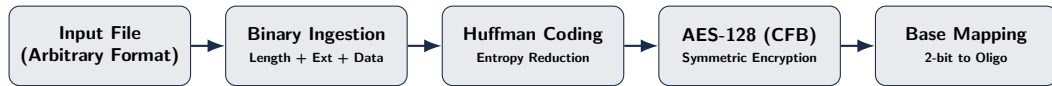
The Confidentiality Vulnerability

- Synthesized biological strands or public sequencers expose unencrypted user data.
- Conventional DNA cryptography literature heavily relies on theoretical One-Time Pad (OTP) models, ignoring Big Data scalability.

Project Objective

Develop an end-to-end, format-agnostic software pipeline incorporating **entropy compression** and **cryptographic security** prior to nucleotide synthesis.

End-to-End Pipeline Architecture



*Strict Order: Compression must precede encryption due to high entropy of ciphertext.

1. Metadata Preservation

Prefixes a 2-byte big-endian header defining extension length and raw extension characters, enabling complete lossless file recovery.

2. The Ordering Rationale

Applying encryption **before** compression eliminates statistical redundancy, destroying any potential compressibility ($H(X) \approx 1$).

Stream Parsing and Entropy Reduction

Algorithmic Implementation

- **Ingestion:** Memory-mapped byte buffers parsed via NumPy unsigned 8-bit arrays (`uint8`).
- **Huffman Encoding:** Constructs an optimal variable-length prefix tree based on payload byte frequencies.
- **Codec Serialization:** Stored via Python pickle (`huff_codec.pkl`) for deterministic decoding.

Biochemical Synthesis Impact

Synthesis Cost Function: $\text{Cost} \propto L_{\text{sequence}}$

By shortening bitstream length via entropy coding, physical synthesis costs and synthesis error windows drop proportionally.

Design Trade-offs

- **Tree Overhead:** Tree metadata must be preserved for reconstruction.
- **Compute Cost:** Calculating global frequency tables introduces an $O(N)$ CPU latency bottleneck on monolithic files.

Compression Ratio

Achieves optimal prefix bounds on structured payloads, providing a minimized footprint before cryptographic expansion.

Cryptographic Design: AES vs. One-Time Pad (OTP)

The Theoretical Fallacy of OTP

- Heavily cited in DNA cryptography for theoretical unbreakability ($H(M|C) = H(M)$).
- **Failure in Big Data Engineering:**
 - Requires true random key length $|K| \geq |M|$.
 - Storing or distributing Petabyte-sized keys doubles the primary storage bottleneck.

Engineering Solution: AES in CFB Mode

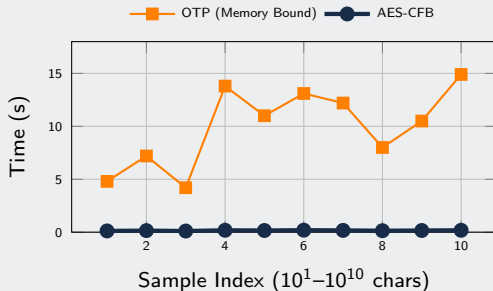
- **Symmetric Security:** Standardized AES-128 with a fixed 16-byte key.
- **Cipher Feedback Mode (CFB):**
 - Converts block cipher into a self-synchronizing stream cipher.
 - Eliminates rigid 16-byte padding requirements.
 - Utilizes a pseudo-random 16-byte IV for semantic security.

Key Management Comparison

Cryptosystem	Key Storage Complexity	Throughput	Applicability to Big Data
One-Time Pad (OTP)	$O(N)$ (Grows with file size)	Memory I/O bound	Impractical for terabyte cold stor
AES-128-CFB	$O(1)$ (Constant 16 bytes)	Hardware Accelerated	Production Ready

Empirical Benchmarks: Execution Latency Analysis

Macro Stress Profile (10^1 to 10^{10} Characters)



Scaling Spectrum (10^1 to 10^{10} Characters)

- **Micro Scale (10^1 – 10^4 chars):**
 - Sub-millisecond latency for both methods.
 - Negligible initial context overhead for AES.
- **Mid Scale (10^4 – 10^6 chars):**
 - Linear expansion in OTP memory footprint.
- **Big Data Domain (10^7 – 10^{10} chars):**
 - Catastrophic RAM exhaustion and latency surges in OTP.
 - AES maintains strictly deterministic, sub-second execution via AES.

Nucleotide Mapping and Biochemical Constraints

2-Bit to Nucleotide Mapping

Bit-Pair	Nucleotide Pair	Base Classification
00	AT	Weak H-Bonds (2x)
01	GC	Strong H-Bonds (3x)
10	CG	Strong H-Bonds (3x)
11	TA	Weak H-Bonds (2x)

Direct 2-base mapping balances purine and pyrimidine distributions across synthesized strands.

Physical Biological Constraints

- **GC-Content Equilibrium:** Desired range: 45%–55% to stabilize DNA melting temperature (T_m) during PCR amplification.
- **Biological Error Regime:** Physical synthesis exhibits $\sim 1\%$ indel/substitution rates.

Outer ECC Requirement

Production systems require concatenation with **Reed-Solomon** or **Fountain Codes** to ensure 100% data recovery.

Scalability Bottlenecks: Transition to Distributed Systems

Single-Node In-Memory Bottlenecks

- Current simulator ingests entire files into RAM via monolithic byte buffers.
- Single-threaded Huffman tree construction scales with high memory pressure at terabyte scale.
- Global codec serialization produces single-point recovery dependencies.

Distributed Chunking Paradigm

- Partition large files into fixed-size chunks (e.g., 100-nt payload + 17-nt index/parity header).
- Process chunks via distributed frameworks (**PySpark** / **MapReduce**).
- Parallelize AES-CFB block hashing across distributed worker nodes.

System Throughput Decoupling

Ingestion Pipeline (GB/s) $\gg$ Physical Synthesis Rate (~ 400 B/s)

Scalable distributed partitioning decouples high-speed data ingestion from slow chemical synthesis reactors.

Project Summary

- Designed and verified an end-to-end secure DNA Data Storage pipeline.
- Demonstrated the empirical and architectural superiority of **AES-CFB over OTP** for Big Data regimes.
- Analyzed biochemical synthesis constraints and identified distributed chunking as the scalability pathway.

Thank you for your time. I welcome your questions and feedback.